



Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 20	
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Batch	2025-29
Class	CSE A2
Academic Year & Semester	26-27 , 3rd Sem
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Title of Assignment:	Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.
Practice questions	1. Create a Java application to manage employee records using CRUD operations
Output Screenshots	import java.sql.*;
Solve questions on given link and attach screenshot of leetcode with your login and o/p	import java.util.Scanner;
	public class EmployeeCRUD {
	public static void main(String[] args) {
	String url = "jdbc:mysql://localhost:3306/companydb";
	String username = "root";

```
String password = "root";

try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(url,
username, password);
    Scanner sc = new Scanner(System.in);

    while (true) {
        System.out.println("\n1. Insert");
        System.out.println("2. Display");
        System.out.println("3. Update");
        System.out.println("4. Delete");
        System.out.println("5. Exit");
        System.out.print("Enter choice: ");
        int choice = sc.nextInt();

        if (choice == 1) {
            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Department: ");
```

```
String department = sc.nextLine();

System.out.print("Enter Salary: ");
double salary = sc.nextDouble();

PreparedStatement ps = con.prepareStatement(
    "INSERT INTO employee VALUES (?, ?, ?, ?)"
);

ps.setInt(1, id);
ps.setString(2, name);
ps.setString(3, department);
ps.setDouble(4, salary);
ps.executeUpdate();

System.out.println("Employee inserted successfully.");
}

else if (choice == 2) {
    Statement stmt = con.createStatement();
    ResultSet rs = stmt.executeQuery("SELECT * FROM
employee");

    System.out.println("\nID\tName\tDepartment\tSalary");

    while (rs.next()) {
```

```
        System.out.println(
            rs.getInt("employee_id") + "\t" +
            rs.getString("name") + "\t" +
            rs.getString("department") + "\t\t" +
            rs.getDouble("salary")
        );
    }
}

else if (choice == 3) {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    System.out.print("Enter New Salary: ");
    double salary = sc.nextDouble();

    PreparedStatement ps = con.prepareStatement(
        "UPDATE employee SET salary = ? WHERE
employee_id = ?"
    );

    ps.setDouble(1, salary);
    ps.setInt(2, id);
    ps.executeUpdate();

    System.out.println("Employee updated successfully.");
}
```

```
}
```

```
else if (choice == 4) {
```

```
    System.out.print("Enter Employee ID: ");
```

```
    int id = sc.nextInt();
```

```
    PreparedStatement ps = con.prepareStatement(
```

```
        "DELETE FROM employee WHERE employee_id = ?"
```

```
    );
```

```
    ps.setInt(1, id);
```

```
    ps.executeUpdate();
```

```
    System.out.println("Employee deleted successfully.");
```

```
}
```

```
else if (choice == 5) {
```

```
    System.out.println("Exiting...");
```

```
    break;
```

```
}
```

```
else {
```

```
    System.out.println("Invalid choice.");
```

```
}
```

```
}
```

```

        con.close();
        sc.close();

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}
}

```

2.Create a Java application to manage student records using CRUD operations. Perform Create,Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database

```

import java.sql.*;
import java.util.Scanner;

public class StudentCRUD {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/studentdb";
        String username = "root";
        String password = "root";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url,
username, password);
            Scanner sc = new Scanner(System.in);

```

```
while (true) {  
    System.out.println("\n===== Student Record Management  
=====");  
    System.out.println("1. Create Student");  
    System.out.println("2. Read Students");  
    System.out.println("3. Update Student");  
    System.out.println("4. Delete Student");  
    System.out.println("5. Exit");  
    System.out.print("Enter choice: ");  
    int choice = sc.nextInt();  
  
    if (choice == 1) {  
        System.out.print("Enter Roll No: ");  
        int rollNo = sc.nextInt();  
        sc.nextLine();  
  
        System.out.print("Enter Name: ");  
        String name = sc.nextLine();  
  
        System.out.print("Enter Course: ");  
        String course = sc.nextLine();  
  
        System.out.print("Enter Marks: ");  
        double marks = sc.nextDouble();
```

```
PreparedStatement ps = con.prepareStatement(
    "INSERT INTO student VALUES (?, ?, ?, ?)"
);

ps.setInt(1, rollNo);
ps.setString(2, name);
ps.setString(3, course);
ps.setDouble(4, marks);
ps.executeUpdate();

System.out.println("Student record created
successfully.");
}

else if (choice == 2) {
    Statement stmt = con.createStatement();
    ResultSet rs = stmt.executeQuery("SELECT * FROM
student");

    System.out.println("\nRoll No\tName\tCourse\tMarks");
    System.out.println("-----");

    while (rs.next()) {
        System.out.println(
            rs.getInt("roll_no") + "\t" +
            rs.getString("name") + "\t" +
```

```
        rs.getString("course") + "\t" +  
        rs.getDouble("marks")  
    );  
    }  
}  
  
else if (choice == 3) {  
    System.out.print("Enter Roll No: ");  
    int rollNo = sc.nextInt();  
  
    System.out.print("Enter New Marks: ");  
    double marks = sc.nextDouble();  
  
    PreparedStatement ps = con.prepareStatement(  
        "UPDATE student SET marks = ? WHERE roll_no = ?"  
    );  
  
    ps.setDouble(1, marks);  
    ps.setInt(2, rollNo);  
    ps.executeUpdate();  
  
    System.out.println("Student record updated  
successfully.");  
}  
  
else if (choice == 4) {
```

```
        System.out.print("Enter Roll No: ");
        int rollNo = sc.nextInt();

        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM student WHERE roll_no = ?"
        );

        ps.setInt(1, rollNo);
        ps.executeUpdate();

        System.out.println("Student record deleted
successfully.");
    }

    else if (choice == 5) {
        System.out.println("Exiting...");
        break;
    }

    else {
        System.out.println("Invalid choice.");
    }
}

con.close();
sc.close();
```

```

    } catch (Exception e) {

        System.out.println("Error: " + e.getMessage());

    }

}

}

```

Output Screenshots (if applicable)

```

PS C:\Users\Arpit\OneDrive\Desktop\JDBC_project> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '@AppData\Local\Temp\cp_7wlp6gpmwvady3fk1rt8e3uq.argfile' 'EmployeeCRUD'

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 1
Enter Employee ID: 101
Enter Name: Arpit
Enter Department: CSE
Enter Salary: 80000
Employee inserted successfully.

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 2

ID      Name      Department      Salary
101     Arpit     CSE             80000.0

===== Employee Record Management =====
1. Insert
2. Display
3. Update
4. Delete
5. Exit
Enter choice: 3
Enter Employee ID: 101
Enter New Salary: 100000
Employee updated successfully.

===== Employee Record Management =====

```

```
===== Employee Record Management =====
```

```
1. Insert
```

```
2. Display
```

```
3. Update
```

```
4. Delete
```

```
5. Exit
```

```
Enter choice: 5
```

```
Exiting...
```

```
PS C:\Users\Arpit\OneDrive\Desktop\JDBC_project> |
```

```
===== Student Record Management =====
```

```
1. Create Student
```

```
2. Read Students
```

```
3. Update Student
```

```
4. Delete Student
```

```
5. Exit
```

```
Enter choice: 1
```

```
Enter Roll No: 102
```

```
Enter Name: Rahul
```

```
Enter Course: IT
```

```
Enter Marks: 98
```

```
Student record created successfully.
```

```
===== Student Record Management =====
```

```
1. Create Student
```

```
2. Read Students
```

```
3. Update Student
```

```
4. Delete Student
```

```
5. Exit
```

```
Enter choice: 2
```

Roll No	Name	Course	Marks
101	Arpit	Computer Science	100.0
102	Rahul	IT	98.0

```
===== Student Record Management =====
```

```
1. Create Student
```

```
2. Read Students
```

```
3. Update Student
```

```
4. Delete Student
```

```
5. Exit
```

```
Enter choice: 3
```

```
Enter Roll No: 101
```

```
Enter New Marks: 95
```

```
Student record updated successfully.
```

	<pre> ===== Student Record Management ===== 1. Create Student 2. Read Students 3. Update Student 4. Delete Student 5. Exit Enter choice: 4 Enter Roll No: 102 Student record deleted successfully. ===== Student Record Management ===== 1. Create Student 2. Read Students 3. Update Student 4. Delete Student 5. Exit Enter choice: 5 Exiting... PS C:\Users\Arpit\OneDrive\Desktop\JDBC_project> </pre>
Conclusion	The program successfully manages student records using CRUD operations—Create, Read, Update, and Delete. JDBC is used to connect the Java application with the MySQL database and perform operations on student details such as roll number, name, course, and marks.